

T-030 / R-042 (R-033 step 2): the latitude-union additive energy is

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

Operator target (R-033): bound the off-diagonal four-circle in-plane incidence energy

$$\sum_{\substack{z_i+z_j=z_k+z_l \\ \{i,j\} \neq \{k,l\}}} \mathcal{I}_{ijkl}, \quad \mathcal{I}_{ijkl} = \#\{c_a + c_b = c_c + c_d : c_a \in C_i, c_b \in C_j, c_c \in C_k, c_d \in C_l\},$$

by $N^2 \log^{O(1)} N$ or at least $o(N^{9/4})$ – which lifts the R-040 sphere $O(N^2)$ conjecture to a theorem. This note is step 2: two PROVED structural results that localise the obstruction; it does NOT close the incidence bound.

2. The height-energy bound (proved)

Proposition 1 (height-energy bound). *For L parallel circles with distinct radii at heights z_i ,*

$$E_+(Q) \leq N^2 + 2 \sum_{k,l} \nu_k \nu_l p(z_k + z_l), \quad p(H) = \#\{(i,j) : z_i + z_j = H\}.$$

For uniform $\nu_i = N/L$ this is $E_+ \leq N^2 + 2(N^2/L^2)E_h$, with $E_h = \sum_H p(H)^2$ the HEIGHT additive energy.

Proof. As in R-040, for $M_{xy} \neq 0$, $r(M_{xy}, M_z) \leq 2p(M_z)$ and $\sum_{M_{xy} \neq 0} r(M_{xy}, M_z) \leq \sum_{(i,j) \in P(M_z)} \nu_i \nu_j$, so $\sum_{M_{xy} \neq 0} r^2 \leq 2p(M_z) \sum_{(i,j) \in P(M_z)} \nu_i \nu_j$; the $M_{xy} = 0$ slice is the diagonal $\leq N^2$ (distinct radii). Summing over M_z and using $\sum_{M_z} p(M_z) \sum_{(i,j) \in P(M_z)} \nu_i \nu_j = \sum_{k,l} \nu_k \nu_l p(z_k + z_l)$ gives the claim. $\square$

This recovers both regimes (assert A2): Sidon heights have $E_h \leq 2L^2$ (so $E_+ = O(N^2)$, matching R-041), AP heights have $E_h \asymp L^3$ (so the bound is $O(LN^2)$, matching R-039). The controlling parameter is the *height* additive energy E_h – a 1-D additive-energy object, a genuine dimension reduction of the latitude-union problem.

3. The sum-annulus mechanism

Proposition 2 (disjoint annuli kill the four-circle term). $\mathcal{I}_{ijkl} = \sum_w r^{ij}(w) r^{kl}(w)$, and $r^{ij}(w) \neq 0$ forces $|w| \in [|\rho_i - \rho_j|, \rho_i + \rho_j]$. Hence if the two sum-annuli are disjoint, $\mathcal{I}_{ijkl} = 0$.

Annulus overlap is therefore NECESSARY for $\mathcal{I}_{ijkl} \neq 0$ (it is NOT sufficient: the discrete sum-sets must also coincide – machine A3 exhibits overlapping circles with $\mathcal{I} = 0$). This is the geometric reason *spread* (distinct, well-separated) radii suppress the four-circle energy to $O(N^2)$ (most cross-pair annuli are disjoint), while *equal* radii (a cylinder) put every annulus at $[0, 2\rho]$ – maximal overlap – and produce the linear LN^2 blow-up (R-040 contrast).

4. What stays open (the R-033 core)

The height-energy bound is LOOSE exactly where E_h is large: at AP $L = 7$ the measured $E_+ = 22988$ is $0.26 \times$ the bound 89456 (assert A4). The gap is the in-plane incidence cancellation – the annulus-overlap pruning of Sec. 3 plus the discrete-coincidence sparsity – which neither the Cauchy–Schwarz per-quadruple bound $\mathcal{I}_{ijkl} \leq 3\sqrt{\nu_i \nu_j \nu_k \nu_l}$ nor the height-energy bound captures. Closing it – i.e. proving $\sum_{z_i+z_j=z_k+z_l, \{i,j\} \neq \{k,l\}} \mathcal{I}_{ijkl} \lesssim N^2 \log^{O(1)} N$ (or $o(N^{9/4})$) – is the open circle-incidence problem and lifts the R-040 sphere conjecture to a theorem. **This is the precise remaining R-033 target.**

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

- (α) “*The height-energy bound is just R-040 reparametrised.*” VALID: it is the same $r \leq 2p(M_z)$ step, reorganised to expose E_h . The value is the dimension reduction (1-D height energy as the parameter) and the explicit Sidon/AP interpolation; no new strength over R-040 is claimed (it gives $O(LN^2)$ for AP, not the empirical $O(N^2)$).
- (β) “*Annulus overlap $\Rightarrow \mathcal{I} > 0$.*” REFUTED and corrected (A3): overlap is necessary, not sufficient; the discrete sum-sets must coincide. Only “disjoint $\Rightarrow \mathcal{I} = 0$ ” is claimed.
- (γ) “*Is this a closure of the sphere conjecture / T-030?*” NO: the bound is loose for high E_h (A4, $0.26 \times$); the incidence cancellation is unproved. T-030 OPEN; the sphere $O(N^2)$ stays STRONG EVIDENCE.

Quantitative sanity check (mandatory). *Bound.* $E_+ \leq N^2 + 2 \sum \nu_k \nu_l p(z_k + z_l)$ on all Sidon+AP configs (A1). *Interpolation.* Sidon $E_h/L^2 \approx 2$ flat, AP $E_h/L^2 = 2.1, 3.4, 4.7$ growing (A2). *Mechanism.* disjoint annuli $\Rightarrow \mathcal{I} = 0$, matching pair $\mathcal{I} = 264 > 0$ (A3). *Gap.* AP $L = 7$ $E_+/\text{bound} = 0.26$ (A4). *Consistency.* all $\leq 3N^{9/4}$ (A5). Deterministic exact-integer; 5/5 asserts, exit 0.

6. Result footer

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Result ID:          R-042 (height-energy bound + sum-annulus mechanism;
                    R-033 step 2) / T-030 frontier
Precise statement:  (1) distinct-radius parallel circles =>
                    E_+ <= N^2 + 2 sum_{k,l} nu_k nu_l p(z_k+z_l)
                    = N^2 + 2(N^2/L^2)E_h (uniform nu); recovers Sidon
                    (E_h~L^2 => O(N^2)) and AP (E_h~L^3 => O(LN^2)).
                    (2) I_{ijkl} != 0 requires the sum-annuli to overlap
                    (DISJOINT => I=0; overlap necessary-not-sufficient) --
                    spread radii suppress, equal radii (cylinder) maximise.
Scope:             parallel circles (latitudes); T-030 (Q in S^2;
                    H-NONLATTICE-REMAINDER). Not load-bearing for C_full head.
Evidence grade:    Both structural results PROVED (elementary). NO closure:
                    height-energy bound is LOOSE for high E_h (E_+/bound=0.26
                    at AP L=7); sphere O(N^2) stays STRONG EVIDENCE (R-040);
                    T-030 arbitrary-Q OPEN (record N^{9/4}, R-033).
Reproduction command: python3 codes/vacuum/dr2_t030_height_energy.py
Expected output:   5/5 asserts PASS, exit 0; JSON at claims/
                    B5-BEYOND-LAYER-BOUND/runs/260614-dr2-t030-height-energy/
Falsifier:         a distinct-radius config with E_+ > N^2+2 sum nu_k nu_l
                    p(z_k+z_l) (refutes the bound); disjoint sum-annuli with
                    I_{ijkl} != 0 (refutes the mechanism). Both checked false.
Tier before / after: NONE -- operator decides. No gate/tier/hypothesis flip.
                    B5 stays T7-SCOPE_{admissibility-bounded}; T-030 OPEN.
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No-overclaim statement: This does NOT close the R-033 incidence bound or the sphere $O(N^2)$ conjecture. The height-energy bound is R-040-strength ($O(LN^2)$ for AP); the annulus mechanism is a necessary condition only. The remaining target is the off-diagonal four-circle incidence energy.

Next required action: R-033 core: bound $\sum_{\{z_i+z_j=z_k+z_l, \{i,j\} \neq \{k,l\}\}} I_{ijkl} \lesssim N^2 \text{polylog}(N)$ (or $o(N^{9/4})$) via circle incidences on overlapping sum-annuli $\rightarrow$ lifts the sphere $O(N^2)$ conjecture to a theorem. Likely needs the Aronov-Sharir / Pach-Sharir incidence machinery (a focused research problem).